Homework — 1.4.1 Data Types

Name: ____ Date: _ ***Mark:*** ___ / 20

OCR H446 — set after the 1.4.1 lesson. SHOW ALL WORKING.

Part A — This week's topic (~14 marks)

1. Convert the denary number **156** to 8-bit unsigned binary. **[2]** *(AO2)*

Working space:

2. Convert the binary number `11001010` to hexadecimal. **[2]** *(AO2)*

Working space:

3. Convert the hexadecimal number `6F` to denary. **[2]** *(AO2)*

Working space:

4. Using 8-bit **two's complement**, calculate **52 − 37** by adding the two's complement of 37. Show the binary working and state clearly what happens to the final carry. **[4]** *(AO2)*

Working space:

5. The 8-bit two's complement value `01001100` is shifted using an **arithmetic right shift of 2 places**. Give the resulting bit pattern and state the arithmetic effect. **[2]** *(AO2)*

Working space:

6. A games developer says, *"Real numbers and integers are basically the same, so it doesn't matter which data type I pick."* Explain **one** reason a programmer must choose the correct data type, referring to how each is represented. **[2]** *(AO1)*

Working space:

Part B — Retrieval: keep it fresh (~6 marks)

7. State the role of the **MAR** (Memory Address Register) in the fetch–decode–execute cycle. *(1.1)* **[1]** *(AO1)*

Working space:

8. Describe how **virtual memory** allows a computer to run a program that is larger than the installed RAM. *(1.2)* **[3]** *(AO1)*

Working space:

9. State **one** difference between **lossless** and **lossy** compression, and give a suitable use for each. *(1.3)* **[2]** *(AO1)*

Working space: